UNIT - IV

Masonry Buildings

Syllabus (as given):

Introduction - elastic properties of masonry assemblage - categories of masonry buildings - behaviour of unreinforced and reinforced masonry walls - behaviour of walls - box action and bands - behaviour of infill walls - improving seismic behaviour of masonry buildings - load combinations and permissible stresses - seismic design requirements - lateral load analysis of masonry buildings.

1. INTRODUCTION

Masonry (brick, stone, concrete block, adobe) is the most common construction in India and the developing world - ~ 70% of the housing stock. Masonry is heavy (large inertia force), strong in compression but weak in tension and shear, brittle, and has little energy dissipation. Therefore unreinforced masonry (URM) buildings are the most vulnerable in earthquakes (Latur 1993, Bhuj 2001, Kashmir 2005, Nepal 2015).

- *Typical failure: wall cracking (X-cracks), out-of-plane collapse of walls, separation at corners, roof/gable collapse, falling parapets.*
- *IS codes: IS 1905 (structural use of unreinforced masonry), IS 4326:2013 (earthquake resistant design & construction of buildings), IS 13828:1993 (improving EQ resistance of low-strength masonry buildings), IS 13827 (earthen buildings), IS 1893 (loads), IS 456 (RC bands).*
- *Load-bearing masonry walls serve as both structure and partition, so the layout of walls (symmetry, continuity) controls seismic behaviour.*

Key idea

Basic principle: since masonry cannot be ductile, seismic resistance is obtained by (i) strength (good materials/mortar), (ii) integrity - box action with RC bands and vertical bars, and (iii) regular, simple configuration.

2. ELASTIC PROPERTIES OF MASONRY ASSEMBLAGE

Masonry is a composite of units (brick/block) and mortar - anisotropic, non-homogeneous. Properties depend on unit strength, mortar strength, bond, workmanship, and moisture.

Property	Symbol / value	Remarks
Compressive strength of masonry	f_m (or f'_m) from prism tests	Depends on f_b (unit) and f_{mortar}; IS 1905 gives basic stress tables; $f_m \approx 0.4\text{-}0.6\, f_b$ for typical mortars
Modulus of elasticity	$E_m = 550\, f_m$ (IS 1905); other codes 300-1000 f_m	For brick masonry $E_m \approx$ 2500-6000 MPa; ASCE: 700 f_m (clay), 900 f_m (concrete)

Property	Symbol / value	Remarks
Modulus of rigidity	$G_m = 0.4\ E_m$	Poisson's ratio $\nu \approx 0.25$ (0.15-0.3)
Tensile (flexural) strength	$f_t \approx 0.05$ - $0.15\ f_m$ ($\approx$ 0.1-0.3 MPa)	Depends on bond between mortar and unit; low, variable
Shear strength	$\tau = \tau_0 + \mu\ \sigma_d$ (Coulomb) with $\tau_0 \approx$ 0.1-0.3 MPa, $\mu \approx$ 0.4-0.6	σ_d = compressive stress from dead load; more gravity load -> more shear strength
Unit weight	18-20 kN/m³ (brick), 22-26 (stone)	Heavy -> large seismic force

- Stress-strain curve: non-linear, brittle; peak at strain ≈ 0.002; crushing 0.003-0.0035. Mortar is usually weaker than unit and expands laterally more -> units are in lateral tension, mortar in triaxial compression -> splitting failure of units.
- Masonry strength is directional: strength normal to bed joints differs from parallel (weakest in the stepped joint direction).
- Prism test (h/t ratio 2-5) gives f_m; slenderness and eccentricity reduce permissible stress by factors (stress reduction factor k_s, area factor k_a, shape factor k_p).
- Cracking occurs at very low strain; stiffness degrades after cracking -> effective stiffness ~ 0.5 of uncracked.

3. CATEGORIES OF MASONRY BUILDINGS

IS 4326:2013 classifies masonry buildings into categories (A, B, C, D, E) based on seismic zone and importance factor I - which then decide the type of strengthening required. Higher category = greater strengthening.

Seismic zone factor / Importance	I = 1.0 (ordinary)	I = 1.5 (important)
II (Z = 0.10)	A	B
III (Z = 0.16)	B	C
IV (Z = 0.24)	C	D
V (Z = 0.36)	D	E

Category from zone and importance (concept as per IS 4326; check table in code)

- Category A - minimum strengthening (e.g., in Zone II); E - most stringent.
- Strengthening items depend on category: horizontal bands (plinth, lintel, roof; sill band), vertical reinforcement at corners/T-junctions/jambs, limits on openings, minimum wall thickness, mortar grade, number of storeys.
- Based on construction type: (i) unreinforced load-bearing masonry, (ii) confined masonry (masonry walls with tie-columns and tie-beams cast after walls), (iii) reinforced masonry (steel in cores/horizontal joints), (iv) infilled RC frames (non-structural masonry in frame), (v) stone/rubble masonry, (vi) adobe/earth.
- Number of storeys is restricted by category and masonry type (IS 4326 Table); storey height ≤ 3.4 m; plinth height ≥ 300-450 mm.

Masonry type	Features	Seismic performance
URM (unreinforced)	Bricks + mortar only	Poor; brittle; avoid in high zones without bands
Confined masonry	Tie columns & beams around walls	Good for low-rise (Chile, Mexico); economical
Reinforced masonry	Steel bars in cores/bed joints, grouted	Good; ductile if detailed (R = 3)
Infilled frame	RC/steel frame + masonry infill	Infill increases stiffness/strength; failure modes (see sec. 7)

4. BEHAVIOUR OF UNREINFORCED & REINFORCED MA

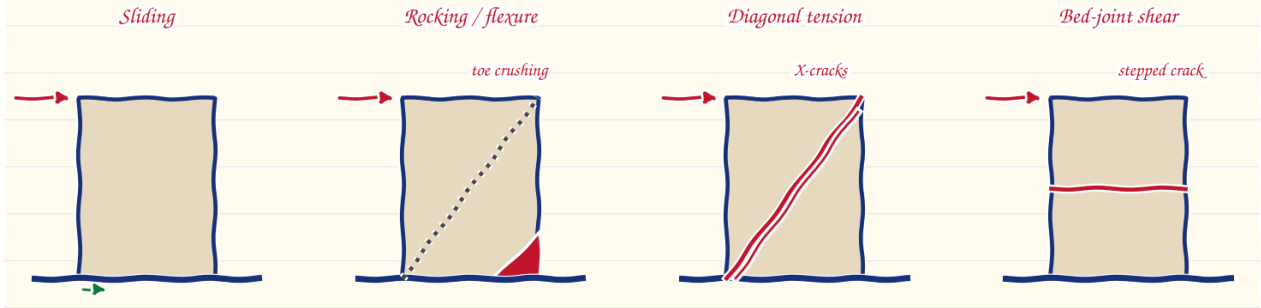


Fig: In-plane failure modes of masonry walls

4.1 In-plane behaviour (shear wall action)

- Bed-joint sliding - horizontal shear crack along a mortar bed joint; occurs at low vertical load / weak mortar.
- Diagonal tension (shear) cracking - X-shaped cracks through units/mortar; most common brittle failure for squat walls with moderate vertical load.
- Rocking (flexure) - slender wall rotates about the toe, tensile cracking at base, toe crushing; ductile compared to shear modes.
- Toe crushing under high vertical load + bending.
- The lateral strength depends on wall length and thickness; the piers between openings act like short columns and fail in shear first. Openings reduce stiffness and strength significantly.

4.2 Out-of-plane behaviour

- Walls perpendicular to ground motion behave as vertical slabs/cantilevers supported at roof and cross-walls; failure by overturning, bending tension cracks mid-height, or horizontal flexural cracking.
- Resistance depends on slenderness ratio (h/t, L/t), edge restraint (cross walls, bands), vertical load; long unsupported walls are most vulnerable (wall length between cross walls ≤ 6 m).
- Gable walls, parapets and cantilevers (chhajjas) fall off first - tie with gable bands and keep parapets ≤ 1 m or reinforce.

4.3 Reinforced masonry

- Vertical bars in cores of hollow blocks or in cavity, horizontal bars in bed joints/bond beams; grout fills the cavities.

- Reinforcement provides tensile and shear capacity, ductility, post-cracking strength and holds fragments together; gives R = 3 (vs 1.5 for URM).
- Typical minimum steel about 0.07% in each direction (total ≈ 0.2%), bars at spacing ≤ 1.2 m, with lap lengths, anchorage and cover as per code (values vary by code).
- Behaviour: flexural cracking -> yielding of vertical steel -> sliding/rocking -> large drift capacity (~ 1-2%) vs ~0.2-0.4% for URM.

Aspect	Unreinforced (URM)	Reinforced masonry
Tension / shear	Almost none (brittle)	Carried by steel
Ductility	Very low, sudden collapse	Moderate; yielding of bars
Drift capacity	0.2 - 0.4%	1 - 2%
R (IS 1893)	1.5	3.0
Post-cracking	Disintegrates, falls	Fragments held together

5. BOX ACTION AND BANDS

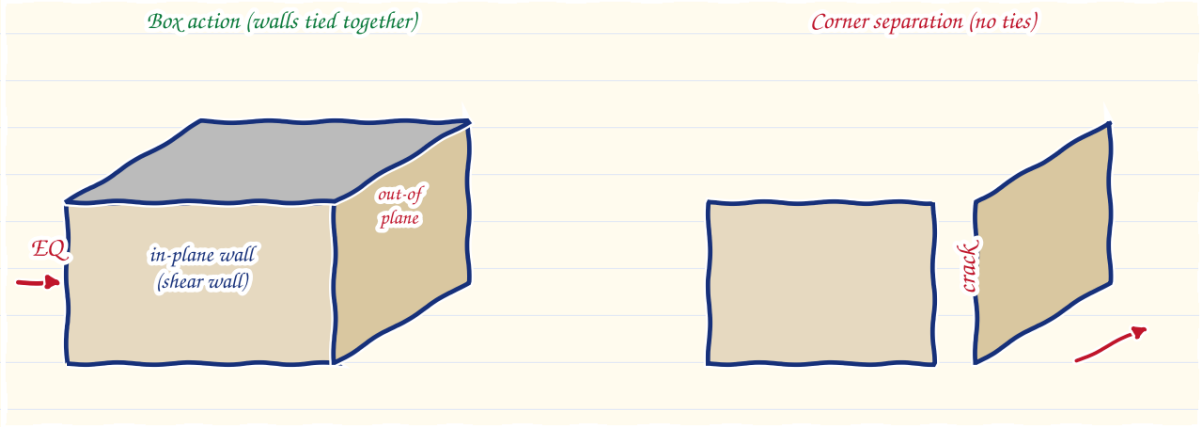


Fig: Box action: walls act together - in-plane walls resist; cross walls hold the out-of-plane walls

- Box action: a masonry building behaves well when all walls and floors/roof are tied together so the building acts like a rigid box - walls parallel to ground motion resist in plane (strong), while walls normal to the motion are supported by them at their ends.
- Without proper connection, corners separate (vertical cracks at corners), walls fall out of plane, roof slides off.
- Requirements: good wall-to-wall bond (toothed joints/stepping), integral horizontal bands, vertical steel at corners/junctions, rigid roof/floor diaphragm (RC slab or tied joists), positive connection of roof to walls.

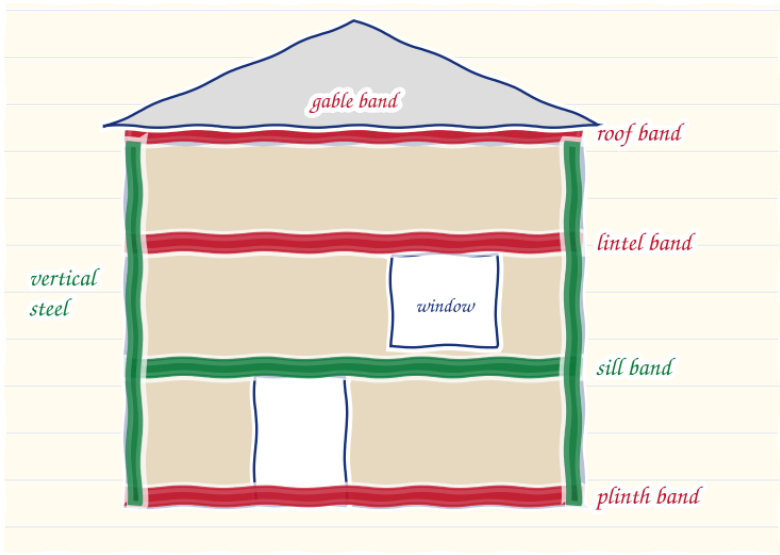


Fig: Location of horizontal seismic bands and vertical steel

Band	Purpose / details
Plinth band	At plinth level in walls on soft/uneven ground; ties walls and reduces differential settlement; not needed if plinth is on firm rock/hard soil with strip footing
Lintel band	At lintel level (all internal and external walls); most important band: ties walls, reduces effective span of walls between cross walls, restrains out-of-plane bending; may be merged with lintels
Roof/floor band	Needed under flexible roofs (timber, sheet) and below RC slab where slab is not properly connected; ensures a continuous tie at the top
Gable band	At the top of gable walls tying them to roof; prevents gable collapse
Sill band	Optional at window sill; in high zones, reduces stress concentration at corners of openings

- RC band: depth ≥ 75 mm (concrete M15/M20 minimum), width = wall thickness; longitudinal steel 2 bars of 8 mm (Category A/B) to 4 bars of 10-12 mm (D/E) with 6 mm stirrups @ 150 mm; continuous around the building; bars properly lapped (bar splices at corners).
- Alternative: timber or steel-bars-in-mortar bands (low-cost buildings, IS 13828).
- Vertical steel at corners, T-junctions, around doors/windows: bar sizes 10-16 mm, anchored in foundation and into roof band; grouted in core of masonry or in "dowels" (increase as category increases).

6. BEHAVIOUR OF INFILL WALLS

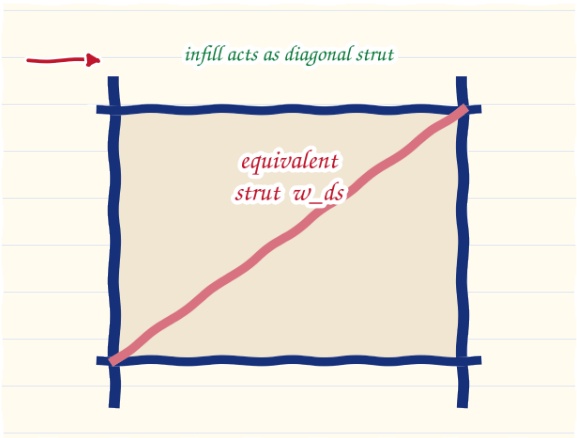


Fig: Infill as an equivalent diagonal strut

- Infill masonry in RC/steel frames is considered non-structural, but it interacts strongly with the frame: it increases stiffness and strength, reduces period and drift initially.
- Under lateral load, frame and infill separate along 2 corners; infill behaves as a compression strut along the diagonal. Equivalent strut width $w_{ds} = 0.175\,\alpha_h^{-0.4}\,L_{ds}$ (IS 1893:2016 cl 7.9.2.2), where $\alpha_h = h\,[E_m\,t\,\sin 2\theta/(4\,E_f\,I_c\,h_{inf})]^{1/4}$; L_{ds} = diagonal length; thickness = infill thickness.
- Positive: extra stiffness/strength/energy dissipation (friction, cracking) - improves performance if present uniformly.
- Negative: (a) infill fails in brittle manner (diagonal cracking, corner crushing); (b) soft storey where ground storey has no infill; (c) short column by partial infill; (d) torsion from asymmetric infill; (e) out-of-plane collapse of slender infill walls; (f) strut action induces high shear in columns -> shear failure.
- Design implications (IS 1893:2016): model infill as struts when stiffness matters; use $T_a = 0.09h/\sqrt{d}$; provide gaps / flexible joints (isolated infill) if frame performance is to be independent; tie infill with dowels/mesh and bands for out-of-plane; avoid openings near columns; ground storey with infill in parking to avoid soft storey.

7. IMPROVING SEISMIC BEHAVIOUR OF MASONRY BUILI

Issue	Provision (IS 4326 / 13828)
Configuration	Simple rectangular plan; L/B ≤ 3; symmetrical; plinth/roof levels uniform; avoid large re-entrant corners; use separate blocks (seismic joint) for L/T plans
Material	Brick strength ≥ 3.5 N/mm² (35 kg/cm²); mortar cement : sand = 1:6 (Cat A,B) to 1:4 (D,E); lime-cement mortar; avoid mud mortar in high zones
Wall thickness & height	t ≥ 200 mm (230 mm brick); storey height ≤ 3.4 m; height/thickness ≤ 12-15; length of wall between cross walls ≤ 6 m
Openings	Total opening length ≤ 50% (1-storey), 42% (2-storey), 33% (3-storey) of wall length; distance from corner ≥ 600 mm; pier width ≥ 600 mm; opening must be centrally located and symmetrical
Bands & steel	Provide plinth, lintel, roof, gable bands; vertical steel at corners & jambs (increase with category)
Bonding	Toothed (stepped) joints at wall intersections; full bond; avoid vertical straight joints; header courses

Issue	Provision (IS 4326 / 13828)
Roof / floor	RC slab preferable; timber/GI roofs braced; tie rafters; roof must act as a rigid diaphragm; no heavy parapets/water tanks on roof
Foundations	Same type/level throughout; strip footings; plinth beam if soil variation
Non-structural	Parapets ≤ 1 m (reinforced), chhajjas ≤ 0.6-1 m; chimneys braced; tie water tanks; avoid cantilevers
Quality	Good workmanship; curing; no loose/eccentric loads

Strengthening

Retrofitting options for existing masonry: ferrocement/mesh jacketing (shotcrete), RC jacketing, horizontal & vertical steel bands (external), grouting of cracks, tying walls (steel rods) at floor level, enlarging piers, adding RC wall/buttress.

8. LOAD COMBINATIONS & PERMISSIBLE STRESSES

- Load combinations (working stress method for masonry as per IS 1905/4326): (i) DL + IL ; (ii) DL + IL ± EL ; (iii) DL ± EL (for overturning/uplift check) ; wind not combined with EQ. IL taken as per IS 1893 (25% / 50% in seismic weight, 0% on roof).
- Permissible stresses under EQ load combination are increased by 33.33% (1/3) over basic values; permissible tensile and shear values depend on mortar and unit strength.
- Basic compressive stress (IS 1905 table) is multiplied by stress-reduction factors for slenderness/eccentricity (k_s), area (k_a) and shape (k_p).
- Permissible shear stress: $\tau_p = \tau_0 + 0.1\,\sigma_d$ (σ_d = dead load stress) - basic shear for mortar M1 (1:6) ~0.05-0.1 N/mm²; min tensile bond ~0.05-0.1 MPa.
- Tension is generally not allowed in URM except small values; where tension develops (overturning), reinforcement must resist it.

9. SEISMIC DESIGN REQUIREMENTS (IS 4326 / 1893)

- Seismic weight W = DL + 25% IL (IL ≤ 3 kN/m²) or 50% IL; roof IL = 0. Seismic zone factor, I, R = 1.5 (URM) to 3.0 (confined/reinforced).
- Design horizontal seismic coefficient $A_h = (Z/2)(S_d/g)/(R/I)$, with S_d/g from IS 1893 spectrum (time period for masonry ≈ 0.09 $h/\sqrt{d}$, usually small, so $S_d/g = 2.5$).
- For low-rise masonry the design coefficient is taken directly from IS 1893 (IS 4326 refers to it); in the older seismic coefficient method $A_h = \beta\, I\, \alpha_0$.
- Vertical projections (parapets, balconies, chhajjas) are designed for about 5 times A_h acting on their weight (IS 1893).
- Walls must be checked for in-plane shear, in-plane flexure, out-of-plane bending (as vertical strip supported at floor/roof levels) and overturning.
- Number of storeys is limited by category and masonry type (IS 4326 Table) - fewer storeys in higher zones, stone/earthen buildings lowest.

10. LATERAL LOAD ANALYSIS OF MASONRY BUILDINGS

1. Compute seismic weight of each floor (roof) including half of wall weights above and below.
2. Base shear $V_B = A_h W$; distribute with height $Q_i = V_B W_i h_i^2 / \Sigma W_j h_j^2$ (for low-rise often triangular).
3. Distribute floor force to walls (in each direction): rigid diaphragm (RC slab) -> in proportion to rigidity (relative stiffness) of walls; flexible diaphragm -> in proportion to tributary mass / area.
4. Add torsion (CM, CR) with accidental eccentricity 0.05 b.
5. Compute shear force in each wall/pier and shear stress $\tau = V/(L\,t)$; compute bending stress and axial stress; check against permissible (+33%).
6. Check overturning / uplift of walls using 0.9 DL.

Wall rigidity (cantilever vs fixed-fixed)

Lateral stiffness of a wall (shear + flexure): $k = 1 / (\Delta \text{ per unit load}) = 1/[\, h^3/(3 E_m I) + 1.2\, h/(G_m A)\,]$ (cantilever)

Fixed-fixed (pier between openings, restrained top & bottom): $\Delta = h^3/(12 E_m I) + 1.2\, h/(G_m A)$

Squat walls ($h/L < 0.4$): shear deformation dominates, $k \approx G_m A/(1.2\, h) = G_m\, t\, L/(1.2\, h)$; slender walls: flexure dominates, $k \propto t\, L^3/h^3$.

Relative rigidity of a wall = $k_{wall} / \Sigma k$ (used for distributing floor shear).

Solved example - single-storey brick building

Data

Single storey, plan 8 m x 6 m, walls 230 mm thick, height 3.0 m, RC slab roof. Zone IV ($Z = 0.24$), $I = 1.0$, $R = 1.5$ (URM), medium soil; walls 19 kN/m³.

Weights: roof slab (150 mm + finish) 5.5 kN/m² x 48 = 264 kN; walls (perimeter 28 m + 1 internal wall 6 m = 34 m): 0.23 x 3.0 x 34 x 19 = 446 kN, half = 223 kN

Seismic weight of roof level: $W = 264 + 223 = 487$ kN (roof LL ignored); $T_a = 0.09 \times 3.0/\sqrt{6} = 0.11$ s -> $S_d/g = 2.5$

$A_h = (0.24/2) \times 2.5/(1.5/1.0) = 0.12 \times 2.5/1.5 = 0.20$; $V_B = 0.20 \times 487 = 97.4$ kN

Distribution to walls in the 6-m direction (loading along 8 m walls)

Two 8 m long walls (no openings) take the shear; assume each rigid by length (squat): V per wall = 97.4/2 = 48.7 kN

Shear stress $\tau = V/(L\,t) = 48.7 \times 10^3/(8000 \times 230) = 0.026$ N/mm²

Permissible shear (mortar 1:6) $\tau_p = 0.05\text{-}0.1 + 0.1\,\sigma_d$; σ_d (roof + wall) ≈ 0.12 N/mm² -> $\tau_p \approx 0.07$ N/mm² x 1.33 = 0.093 N/mm² -> 0.026 < 0.093 OK

Out-of-plane: 6-m wall spanning 8 m wall-to-wall - verify $h/t = 3000/230 = 13$ (limit ~ 15) and add lintel band + vertical steel at corners.

11. MORE WORKED EXAMPLES

11.1 Relative rigidity of walls with openings

Wall with two piers (cantilever method)

Wall height $h = 3.0$ m, thickness $t = 0.23$ m, $E_m = 2750$ N/mm², $G_m = 0.4\,E_m = 1100$ N/mm². Piers: P1 length 1.2 m, P2 length 2.4 m.

Cantilever stiffness: $k = 1/[h^3/(3E_m I) + 1.2h/(G_m A)]$ with $I = t L^3/12$, $A = tL$

P1: $I = 0.23 \times 1.2^3/12 = 0.0331$ m⁴ -> flexure $27/(3 \times 2.75e6 \times 0.0331) = 9.9e\text{-}5$ m/kN; shear $3.6/(1.1e6 \times 0.276) = 1.19e\text{-}5$ -> $\Delta = 1.11e\text{-}4$ -> $k_1 = 9000$ kN/m

P2: $I = 0.23 \times 2.4^3/12 = 0.265$ m⁴ -> flexure $27/(3 \times 2.75e6 \times 0.265) = 1.23e\text{-}5$; shear $3.6/(1.1e6 \times 0.552) = 5.9e\text{-}6$ -> $\Delta = 1.82e\text{-}5$ -> $k_2 = 54\,900$ kN/m

Share of storey shear: P1 = 9000/63 900 = 14%, P2 = 86% -> longer pier attracts most force; narrow piers rely on bands/steel.

Learning

Observation: stiffness $\propto L^3$ for slender walls but $\propto L$ for squat walls. Door/window openings cut the wall into piers and reduce lateral strength drastically - hence opening limits (50%/42%/33%).

11.2 Overturning check

Check wall stability

Wall 4 m long, 3 m high, $t = 0.23$ m, with roof load 30 kN/m on top; base shear force at roof $V = 40$ kN (A_h W portion).

Weight $W = 0.23 \times 3 \times 4 \times 19 = 52.4$ kN + roof $30 \times 4 = 120$ kN -> $N = 172.4$ kN; use 0.9 DL = 155 kN for overturning

Overturning moment $M = 40 \times 3 = 120$ kN.m; restoring moment $= 155 \times 4/2 = 310$ kN.m -> F.S. = 2.6 > 1.5 OK

Base stress $\sigma = N/A \pm M/Z = 155/(0.23x4) \pm 120/(0.23 \times 4^2/6) = 168 \pm 196$ (kN/m²) -> tension 28 kN/m² at heel: small tension -> reinforce vertical bars / increase dead load.

11.3 Out-of-plane check of a wall strip

- Treat a 1 m strip as vertical slab supported at base and roof; seismic pressure $p = A_h \times w$ (w = weight per m² of wall = $0.23 \times 19 = 4.4$ kN/m²).
- For $A_h = 0.20$: $p = 0.88$ kN/m²; moment (propped cantilever / simply supported) $M = p\,h^2/8 = 0.88 \times 9/8 = 0.99$ kN.m/m
- Section modulus $Z = 1 \times 0.23^2/6 = 8.8 \times 10^{-3}$ m³ -> bending stress $\sigma = 0.99/8.8e\text{-}3 = 112$ kN/m² (0.11 MPa) vs permissible tensile strength of masonry about 0.05-0.1 MPa -> cracks; need vertical steel/band or reduce slenderness.

12. STONE, EARTHEN AND LOW-STRENGTH MASONRY

- Stone masonry: heavy, irregular, weak mortar (mud/lime) -> poor performance (Latur 1993). Needs through-stones (one per 600 mm), horizontal bands (timber/RC), vertical bars at corners (IS

4020, IS 13828). Limit: height ≤ 1-2 storeys, thickness ≥ 450 mm.

- Adobe / earth buildings (IS 13827): very heavy and weak; strengthened by bamboo/steel mesh, wall-corner reinforcement, light roof (reduce mass), buttresses, strong plinth.
- Low-strength masonry (IS 13828): brick in mud mortar - provide rope-like bands at plinth, lintel, roof; vertical bars at corners; roof tied by a ring beam; use light roof (GI sheet/tiles instead of heavy earth roof).
- Common local techniques: bamboo reinforcement, dhajji dewari (timber frame with infill - Kashmir), Bhatar (Himachal timber-laced stone masonry).

Confined masonry (modern low-rise practice)

- Masonry walls are built first; then RC tie-columns at corners/jambs and tie-beams at floor levels are cast against the walls (toothed interface). Walls carry the vertical load, the frame confines them.
- Confining elements prevent wall disintegration after cracking, raise drift capacity (~ 0.5-1%), and give box action; limits: tie-columns ≥ 150 mm (typ. 4 bars 10 mm), tie-beam at each floor, spacing of columns ≤ 4-5 m.

13. LESSONS FROM PAST EARTHQUAKES - MASONRY

Earthquake	Observed masonry damage	Lesson
Latur 1993 (M 6.2)	Stone masonry with heavy mud roofs collapsed; ~8000 deaths	Light roof, bands, vertical steel, avoid mud mortar
Bhuj 2001 (M 7.7)	Brick/stone houses collapsed, corner separation, parapets and gables fell	Box action, bands, quality, anchorage of parapets
Kashmir 2005 (M 7.6)	Stone masonry with heavy roofs; dhajji dewari survived better	Light timber-braced systems good
Nepal 2015 (M 7.8)	Mud-mortar stone & brick URM; temples; infilled RC soft-storey	Retrofitting and training of masons
Sikkim 2011	Infill and masonry damage, poor detailing	Seismic code compliance

Common deficiencies in masonry buildings

- Poor mortar and workmanship; no bands; unbraced long walls; large openings near corners; heavy RC/earth roofs on weak walls; shallow/unequal foundations; unreinforced parapets and gables; cut-out walls for services.
- Inadequate roof-to-wall connection - roof slides off walls; floor joists with no anchorage.

FORMULA SHEET - UNIT IV

Item	Formula / value
Elastic modulus	$E_m = 550\, f_m$; $G_m = 0.4\, E_m$; $\nu \approx 0.25$
Shear strength	$\tau = \tau_0 + \mu\, \sigma_d$ (permissible: $\tau_0 + 0.1\, \sigma_d$, $\times 1.33$ for EQ)
Design coefficient	$A_h = (Z/2)(S_a/g)/(R/I)$; $R = 1.5$ (URM), 3.0 (reinforced)

Item	Formula / value
Base shear and distribution	$V_B = A_h W$; $Q_i = V_B W_i h_i^2 / \Sigma W h^2$
Wall stiffness	$k = 1/[h^3/(3E_m I) + 1.2h/(G_m A)]$ (cantilever)
Shear stress in wall	$\tau = V/(L t)$
Openings limit	≤ 50% / 42% / 33% (1 / 2 / 3 storeys); corner distance ≥ 600 mm
Infill strut	$w_{ds} = 0.175 \alpha_h^{-0.4} L_{ds}$

SHORT ANSWERS (2-mark / viva)

Question	Answer
Why is masonry vulnerable?	Heavy (high inertia), brittle, very low tensile & shear strength, little ductility, poor connections
What is box action?	Walls, floors and roof tied so that the building responds as a single rigid box; in-plane walls resist, cross walls support out-of-plane walls
Which band is most important?	Lintel band (ties all walls, reduces out-of-plane bending)
Role of vertical steel?	Resists vertical bending/rocking, holds corners, increases ductility and prevents collapse of wall pieces
Max length/height of unsupported wall?	Wall length between cross walls ≤ 6 m; storey height ≤ 3.4 m; h/t ≤ 12-15
Why limit openings?	Openings weaken wall; piers fail in shear; limit 50% / 42% / 33% of wall length
Infill: advantage and disadvantage?	Adds stiffness/strength; but creates soft storey, short column, torsion; brittle failure
How is mortar strength specified?	By cement : sand ratio (1:6 to 1:4) or compressive strength grades M1-M3 (IS 1905)
R for URM / reinforced masonry?	1.5 / 3.0 (IS 1893:2016)
How to improve out-of-plane strength?	Reduce slenderness, add cross walls/bands/buttresses, vertical steel, tie wall to roof

Checklist for a seismic-safe masonry house (IS 4326)

1. Simple rectangular plan, L/B ≤ 3, symmetric; separate blocks if irregular.
2. Good bricks/stones, richer mortar, full bond with toothed corners.
3. RC plinth band (if needed), lintel band, roof band, gable band - continuous & lapped properly.
4. Vertical bars at all corners/junctions/jambs, anchored in foundation and roof band.
5. Openings small, central and ≥ 600 mm from corners; total ≤ 50% (1 storey).
6. Light roof / RC slab tied to walls; no heavy parapets/overhangs/water tanks on top.
7. Uniform foundation level and type; good drainage; quality workmanship and curing.

QUICK REVISION - Likely questions

1. Elastic properties of masonry: E_m, G_m, strength, stress-strain behaviour.
2. Categories of masonry buildings (A-E) as per IS 4326; strengthening required.
3. Behaviour of URM and reinforced masonry walls in-plane and out-of-plane; failure modes.
4. Explain box action; role of horizontal bands and vertical reinforcement (sketch).
5. Behaviour of infill walls, equivalent strut, positive and negative effects.
6. Measures to improve seismic behaviour of masonry buildings (IS 4326 provisions).
7. Load combinations/permissible stresses; steps of lateral load analysis; numerical on base shear and wall shear.